

IS5411 画图大全

把 php 听成 PhD 的玥姐

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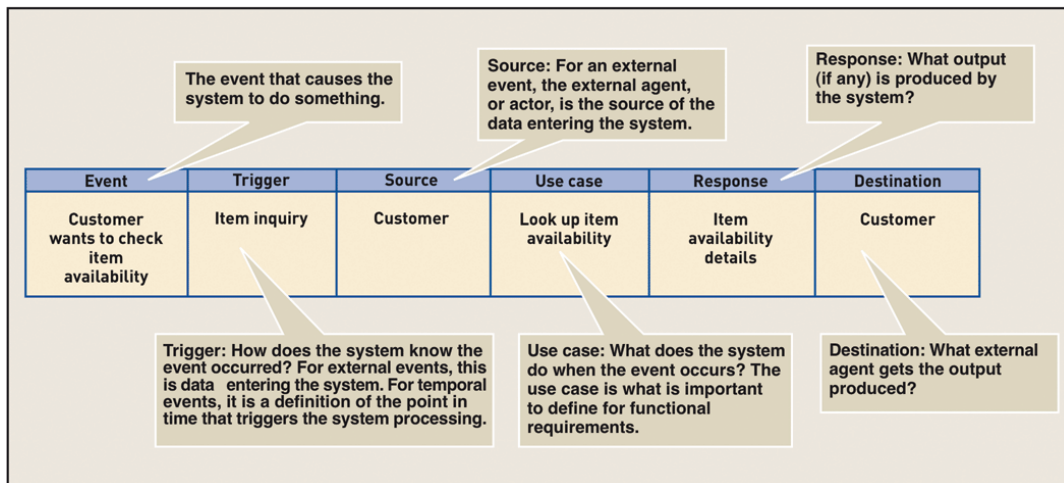
Use case

识别：动词+名词，一时一地&有意义，用户目标+状态改变+事件分步

例：Login system 不是 use case, register system 是 use case

Event Table

按事件列出的 use case，包括一些细节



Customer support system event table					
Event	Trigger	Source	Use case	Response	Destination
1. Customer wants to check item availability	Item inquiry	Customer	Look up item availability	Item availability details	Customer
2. Customer places an order	New order	Customer	Create new order	Real-time link	Credit bureau
				Order confirmation	Customer
				Order details	Shipping
				Transaction	Bank
3. Customer changes or cancels order	Order change request	Customer	Update order	Change confirmation	Customer
				Order change details	Shipping
				Transaction	Bank
4. Time to produce order summary reports	"End of week, month, quarter and year"		Produce order summary reports	Order summary reports	Management
5. Time to produce transaction summary reports	"End of day"		Produce transaction summary reports	Transaction summary reports	Accounting
6. Customer or management wants to check order status	Order status inquiry	Customer or management	Look up order status	Order status details	Customer or management

Use Case Description

描述具体步骤

Brief: 一段话

Create new order description

When the customer calls to order, the order clerk and system verify customer information, create a new order, add items to the order, verify payment, create the order transaction, and finalize the order.

Intermediate: 中等描述

Flow of activities for scenario of *Order Clerk creates telephone order*

Main Flow:

1. Customer calls RMO and gets order clerk.
2. Order clerk verifies customer information. If a new customer, invoke *Maintain customer account information* use case to add a new customer.
3. Clerk initiates the creation of a new order.
4. Customer requests an item be added to the order.
5. Clerk verifies the item and adds it to the order.
6. Repeat steps 4 and 5 until all items are added to the order.
7. Customer indicates end of order; clerk enters end of order; system computes totals.
8. Customer submits payment; clerk enters amount; system verifies payment.
9. System finalizes order.

Exception Conditions:

5. If an item is not in stock, then customer can
 - a. choose not to purchase item, or
 - b. request item be added as a back-ordered item.
8. If customer payment is rejected due to bad-credit verification, then
 - a. order is canceled, or
 - b. order is put on hold until check is received.

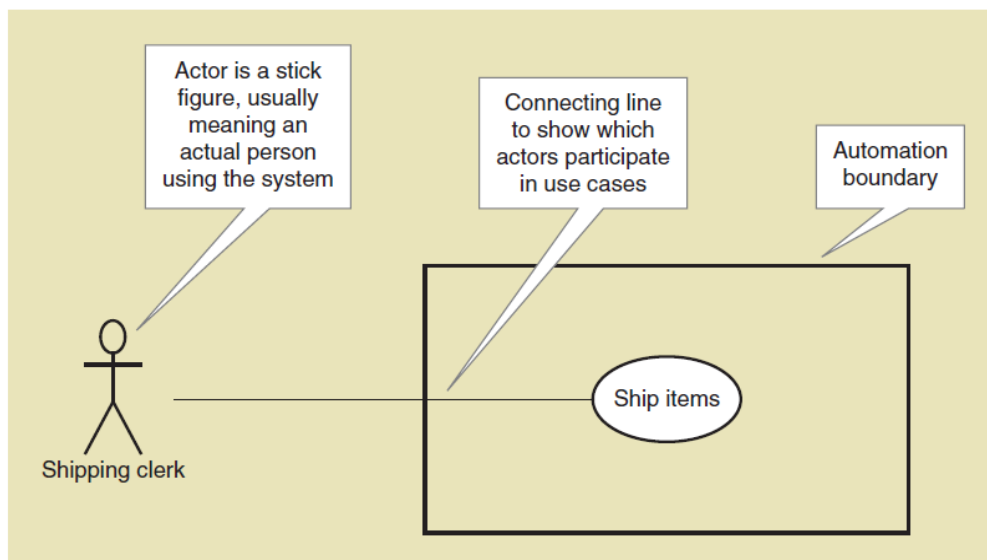
Fully Developed Description——IS5411 不要求

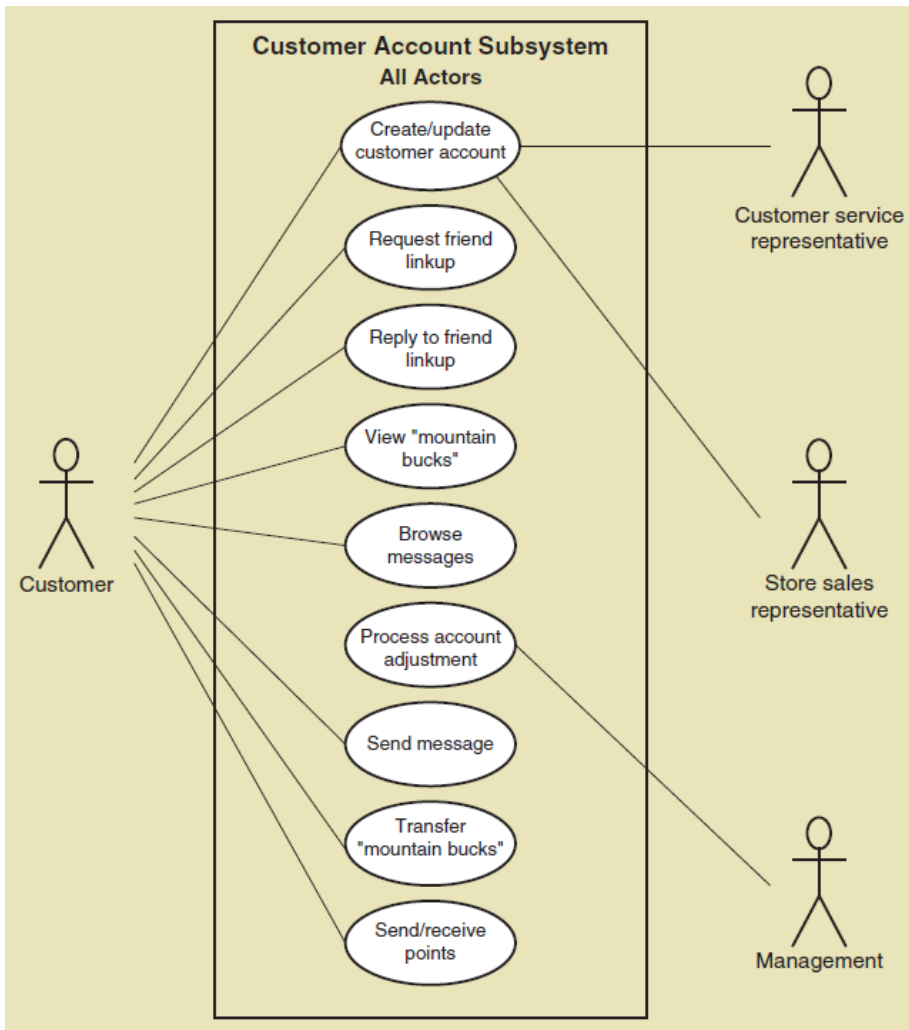
Use Case Diagram

遵循 UML 规则 (Unified Modelling Language)

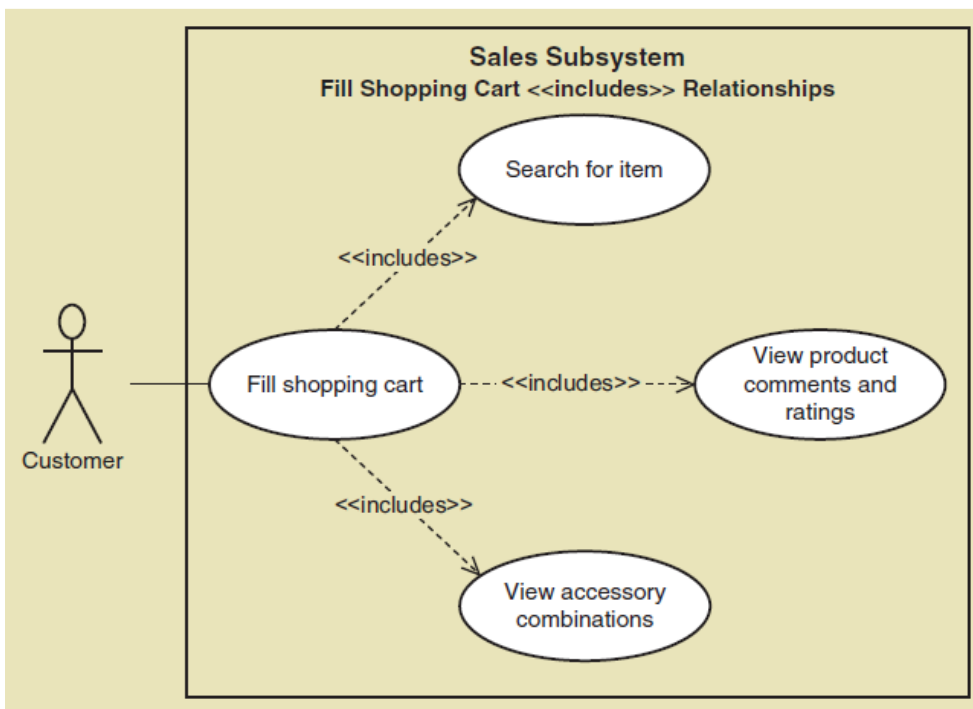
题目里会出现 UML，只是一套画图规则罢了，不特定指代某一种图

有人 **end user**、有圈 **use case**、有框 **automation boundary**、连线（人和圈）





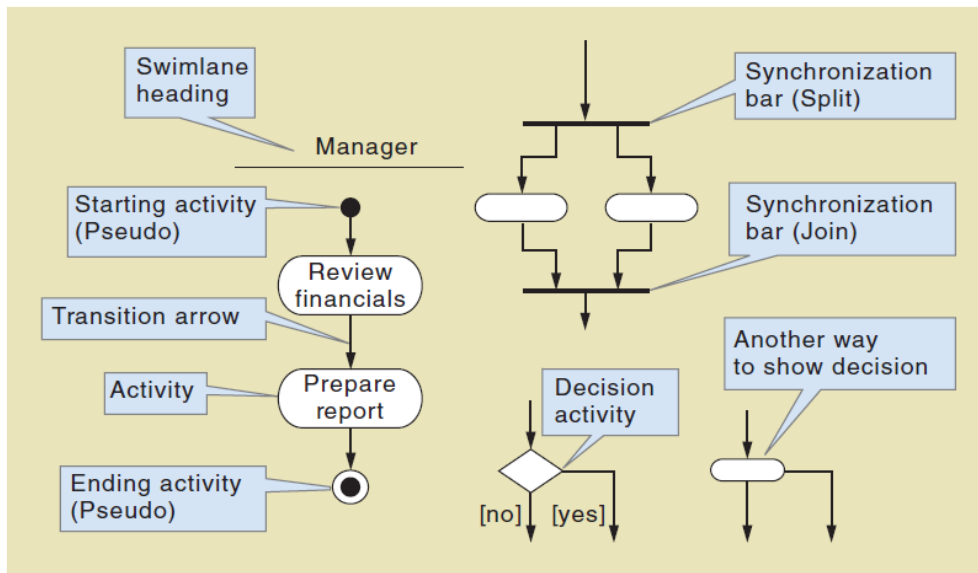
The <<Includes>> relationship 包含关系——不考，看看算了



虚线箭头上写<<Includes>>

Activity Diagram

Workflow 工作流程，考试一般就画 2、3 列的



开始：黑点。结束：黑带外面套白圈；步骤：圆角方框

LOOP 循环：两个黑色杠之间，上下箭头都是一个

第一个杠上写 For each item, 第二杠上写 End for each

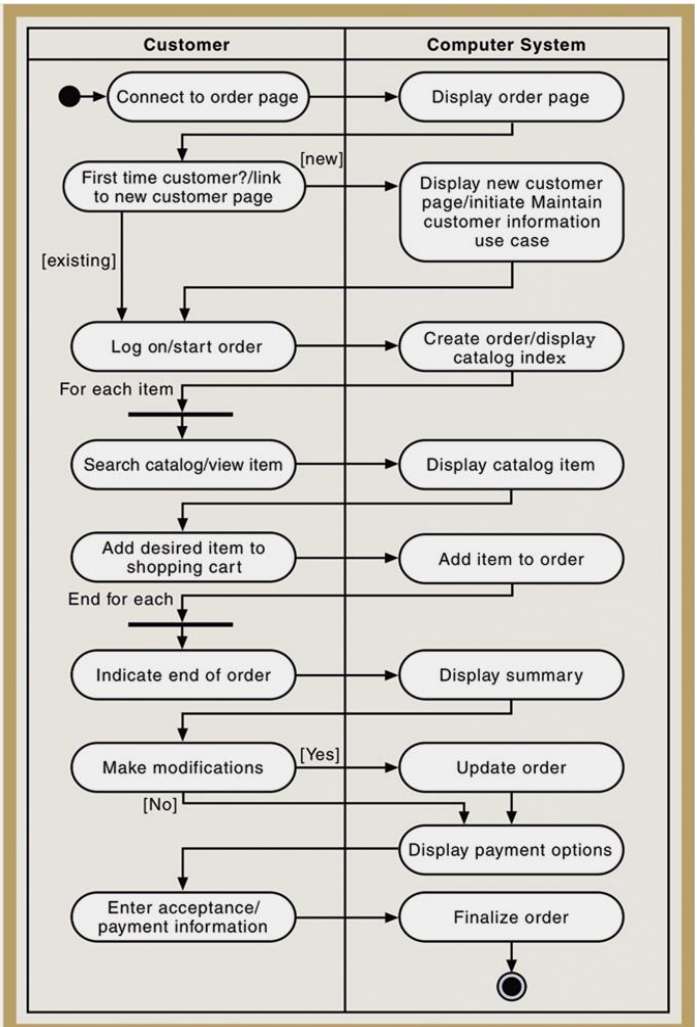
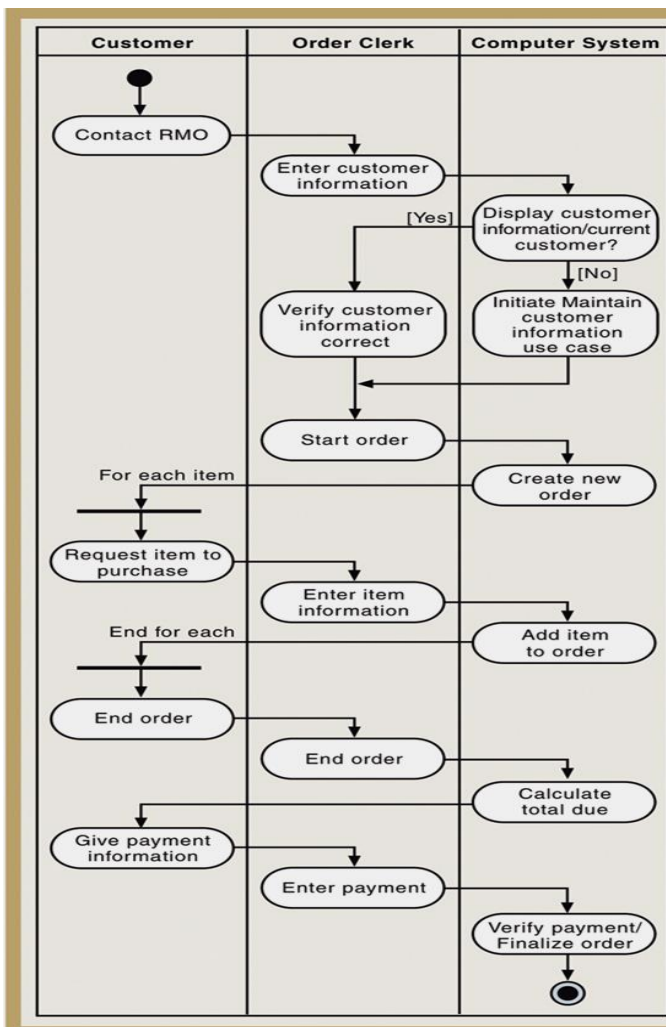
分两条路（1 分 2 或 2 合 1）：黑色杠，上下箭头一单一双

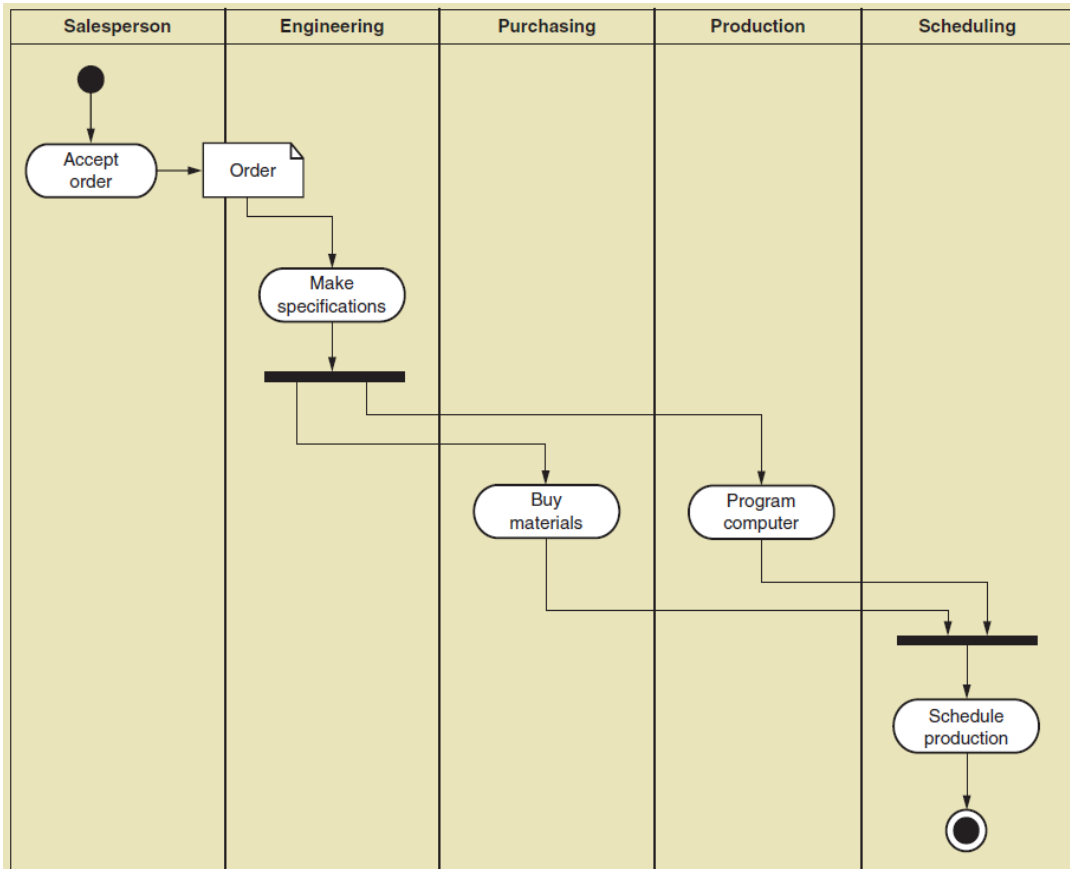
判断是否：菱形，两条线，每条标明[yes]或[no], [existing]或[new]

Alternative 表示方法：圆角方框，两条线，在[]里标明条件

判断：在相应的流程线旁边画[], 里面写条件（yes & no, existing & new 等）

循环：开始在黑杠上方直接写, For each XXX; 结束也在黑杠上方直接写, End for each





Noun Table

想&找名词，再辨别：是 **class** 还是 **attribute** 或者啥也不是

常见：角色、功能、流程、系统、报告、类别

删除近义词、识别是不是其他词的一部分、识别输入&输出

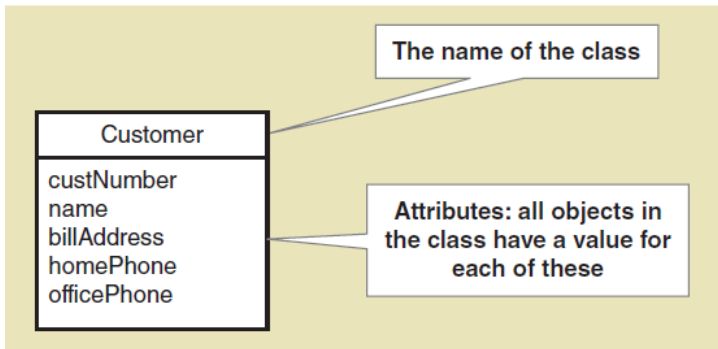
Identified noun	Notes on including noun as a thing to store
Accounting	We know who they are. No need to store it.
Back order	A special type of order? Or a value of order status? Research.
Back-order information	An output that can be produced from other information.
Bank	Only one of them. No need to store.
Catalog	Yes, need to recall them, for different seasons and years. Include.
Catalog activity reports	An output that can be produced from other information. Not stored.
Catalog details	Same as catalog? Or the same as product items in the catalog? Research.
Change request	An input resulting in remembering changes to an order.
Charge adjustment	An input resulting in a transaction.
Color	One piece of information about a product item.
Confirmation	An output produced from other information. Not stored.
Credit card information	Part of an order? Or part of customer information? Research.
Customer	Yes, a key thing with lots of details required. Include.
Customer account	Possibly required if an RMO payment plan is included. Research.
Fulfillment reports	An output produced from information about shipments. Not stored.
Inventory quantity	One piece of information about a product item. Research.
Management	We know who they are. No need to store.
Marketing	We know who they are. No need to store.
Merchandising	We know who they are. No need to store.

（这个应该不考画图）

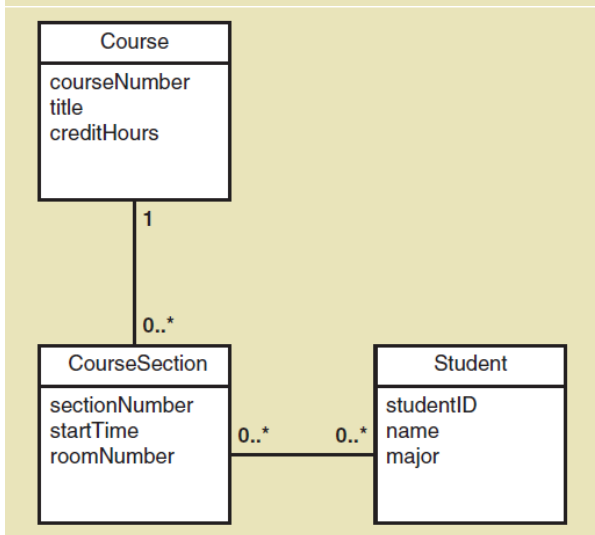
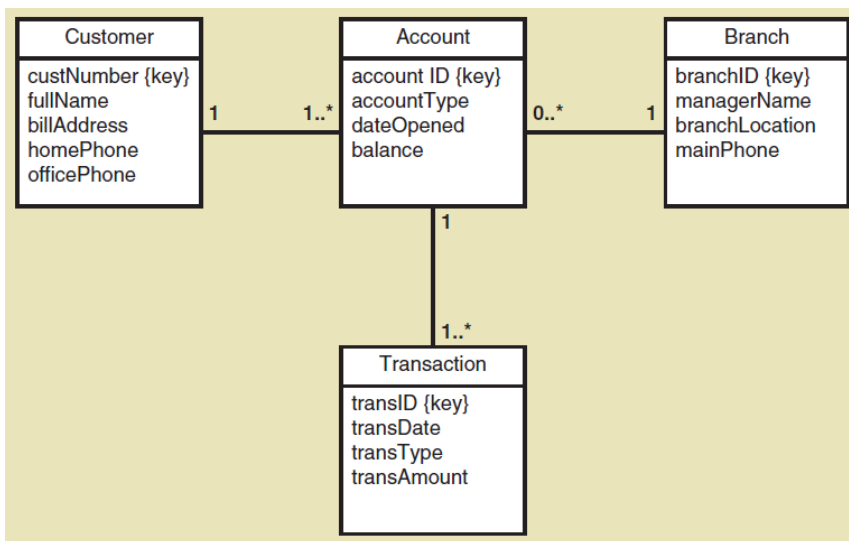
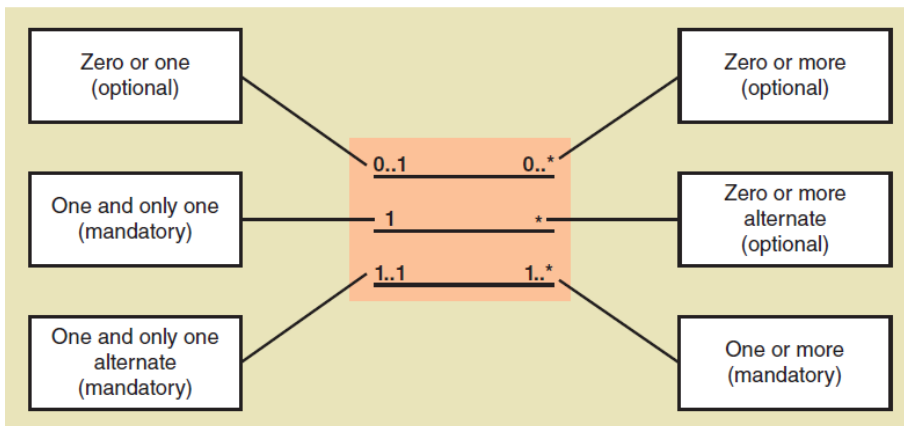
Class Diagram

类的 Title 所有单词首字母都大写，不空格

Attribute 遵循 camelback notation——第一个单词小写，第二个及之后单词首字母大写，都不空格

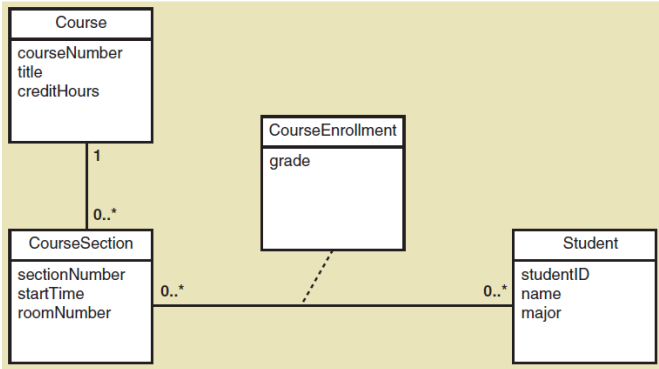


1 关系——挨着谁近就是说谁的数量
 至少是 0，关系可选
 至少是 1，关系必须



1. 相关关系

Association Class 关联类——多对多关系中才有，虚线链接，跟俩 class 都有关系



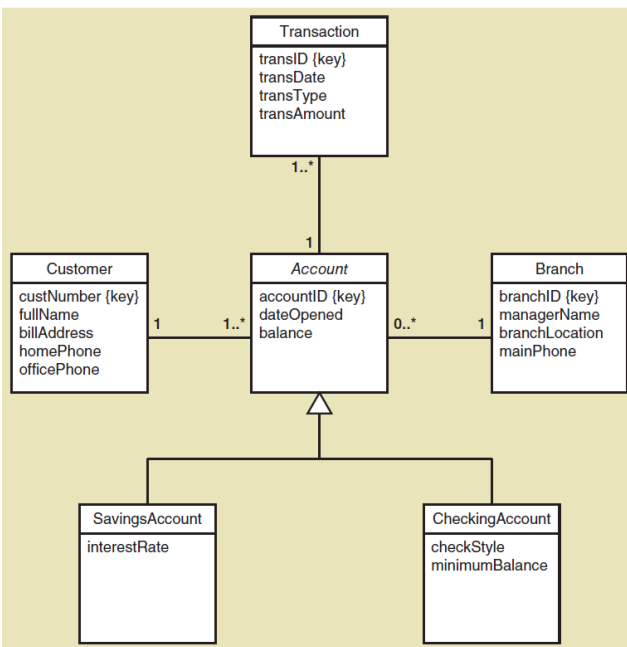
2. 继承关系：上下级、子类

空心三角箭头表示子类，指向上级

A SavingsAccount has 4 attributes (=3+1)

A CheckingAccount Has 5 attributes (=3+2)

属性与关系都继承

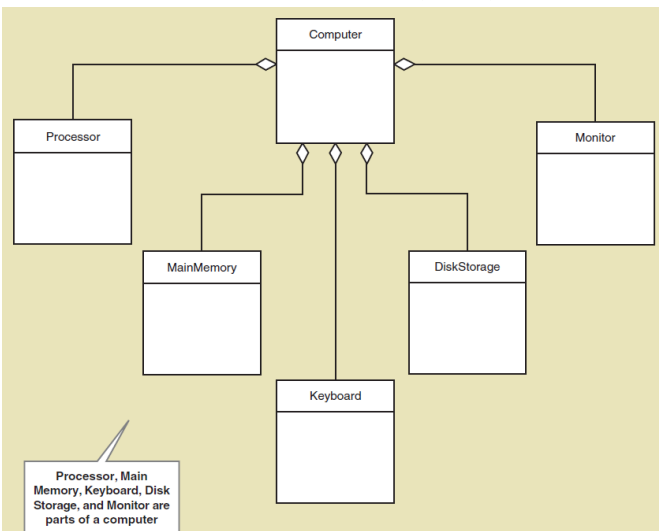


3. 部分整体关系：组成部分

菱形表示

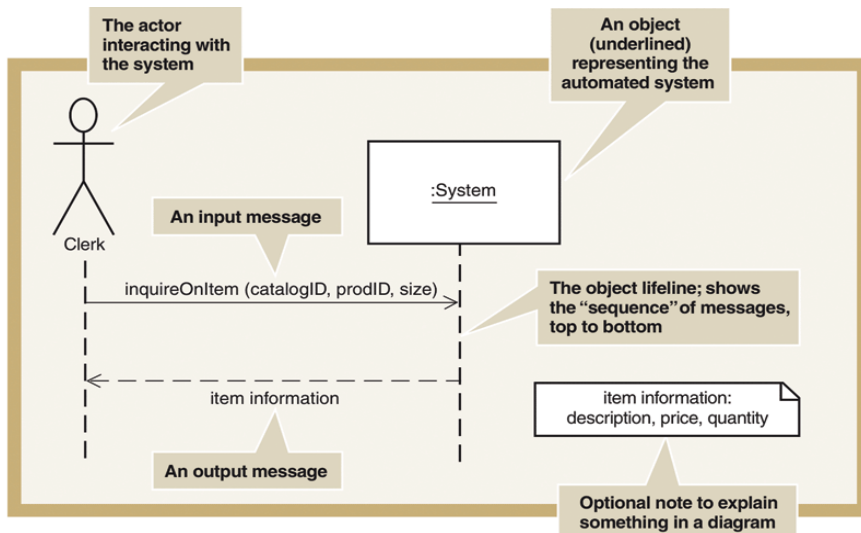
聚集——部分可以被删除和替换，比如汽车的轮子、电脑的硬盘

必要组成——部分不可以被删除和替换，比如人的手指



System Sequence Diagram (SSD)

包括输入输出

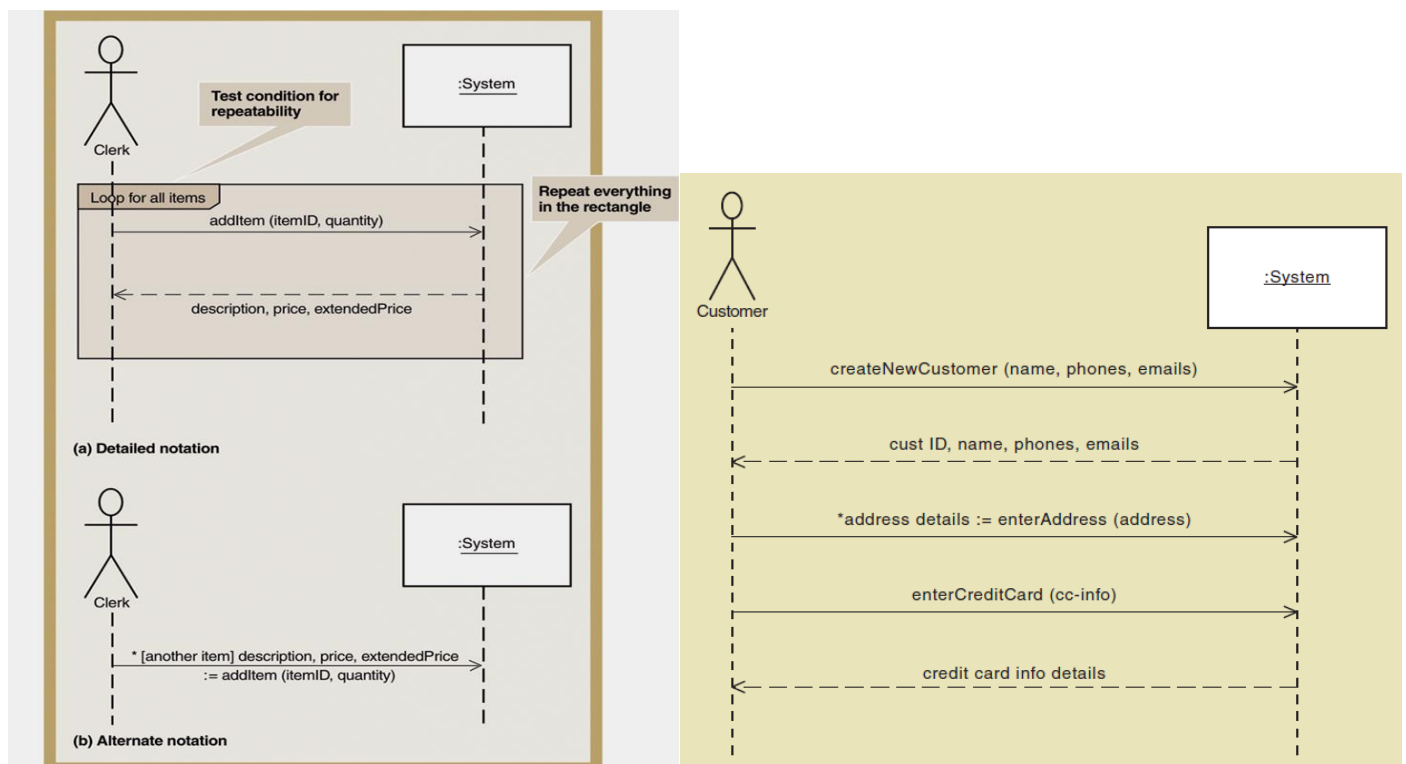


冒号前：命名。冒号后：实体类别（是个啥东西）

下划线：自动系统，输入实线，输出虚线，两侧虚线

文字：动词小写，类(class)的每个单词首字母都大写，不空格。不是类(class)的其他东西都小写

Repeating Message——loop 框



常规画法：

Loop 方框框起来，左上角标注 Loop for XXX

输入实线，输出虚线

简便画法：

一条实线，输入输出共用

线上输出，线下输入，实线下方的输入前需添加**:=**

或者:=前输出，:=后输入，都在线上

[]里是条件，*****代表 loop，***后[]**里的条件满足，才开始循环

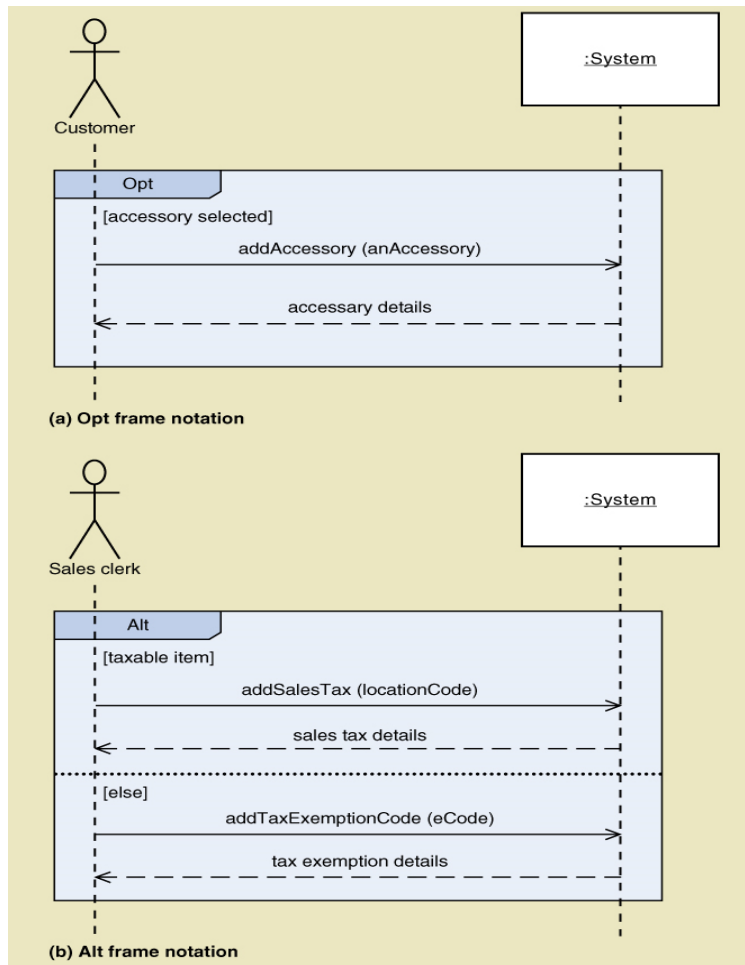
简便画法中只有*，没有 loop 框，常规画法才有 loop 框

SSD 大小写：attribute 都小写；和命令动词在一起的 item 或 attribute，动词小写，item 或 attribute 首字母大写，都不空格

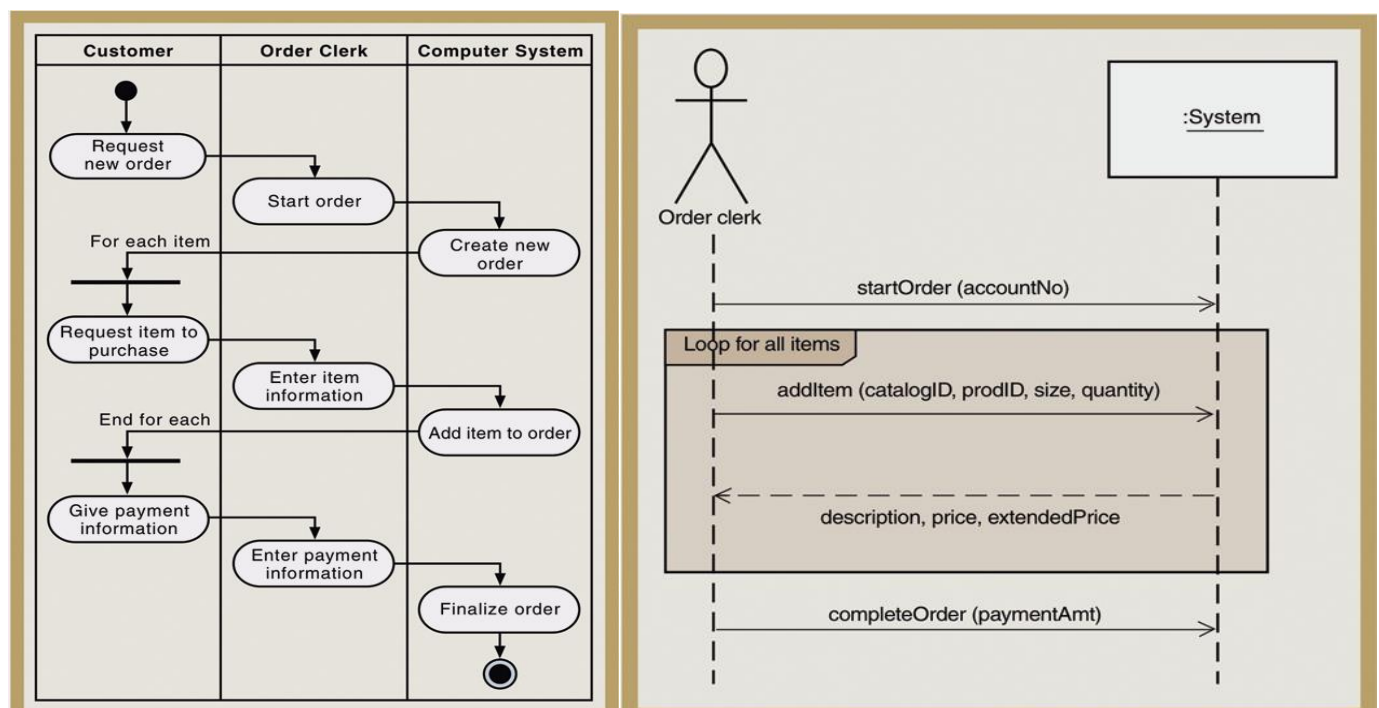
输入一般动词+名词(), 如果需要检索, ()内需要添加相应 attribute (比如 ID, name 之类的)
 输出不要命令, 只要 attribute 即可

Opt Frame (optional) **选择**——框里直接写 Opt, 表示是否, 0 or 1。需要在[]内注明条件

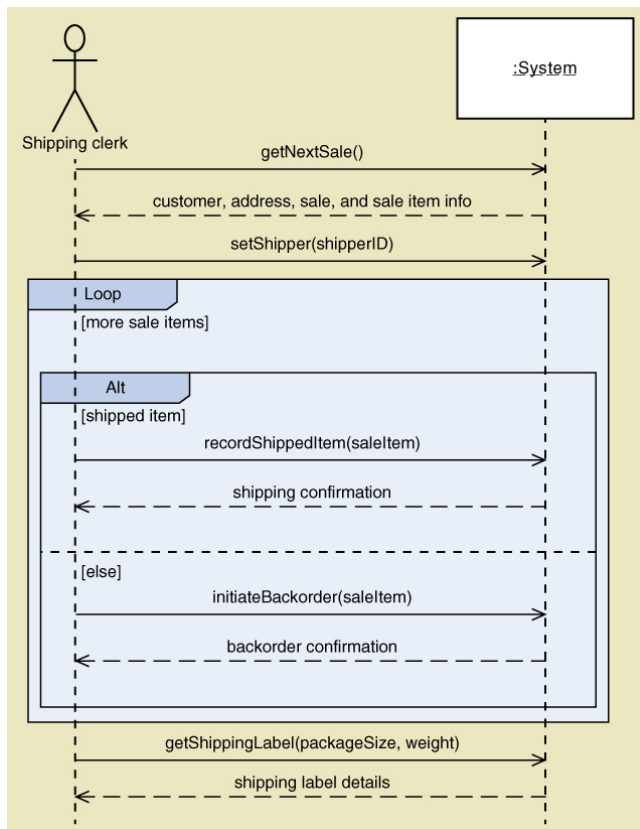
Alt Frame (if-else, alternative) **条件**——框里直接写 Alt, 表示 N 选 1, N 也可以等于 2。在[]内注明条件。中间横线隔开, 虚线实线都可以, 隔开的线下方[else]。一个小框里, 最好输入&输出成对出现。
 具体解释参照 (这个网友讲挺好): <http://www.cnblogs.com/cy568searchx/p/6227238.html>



例子: 根据 activity diagram (workflow) 画 SSD



例子 2: 看个复杂的, 大的套小的



(这个图看会, 考试 SSD 无忧)

First-Cut Design Class Diagram (First-Cut DCD)

需要添加<<controller>>这个 class, 放在第一步

XXXHandler, 这个 XXX 就是 use case 的名字或者动词, 每个单词首字母都大写不空格

所有 object 的每个单词的首字母大写, 不空格

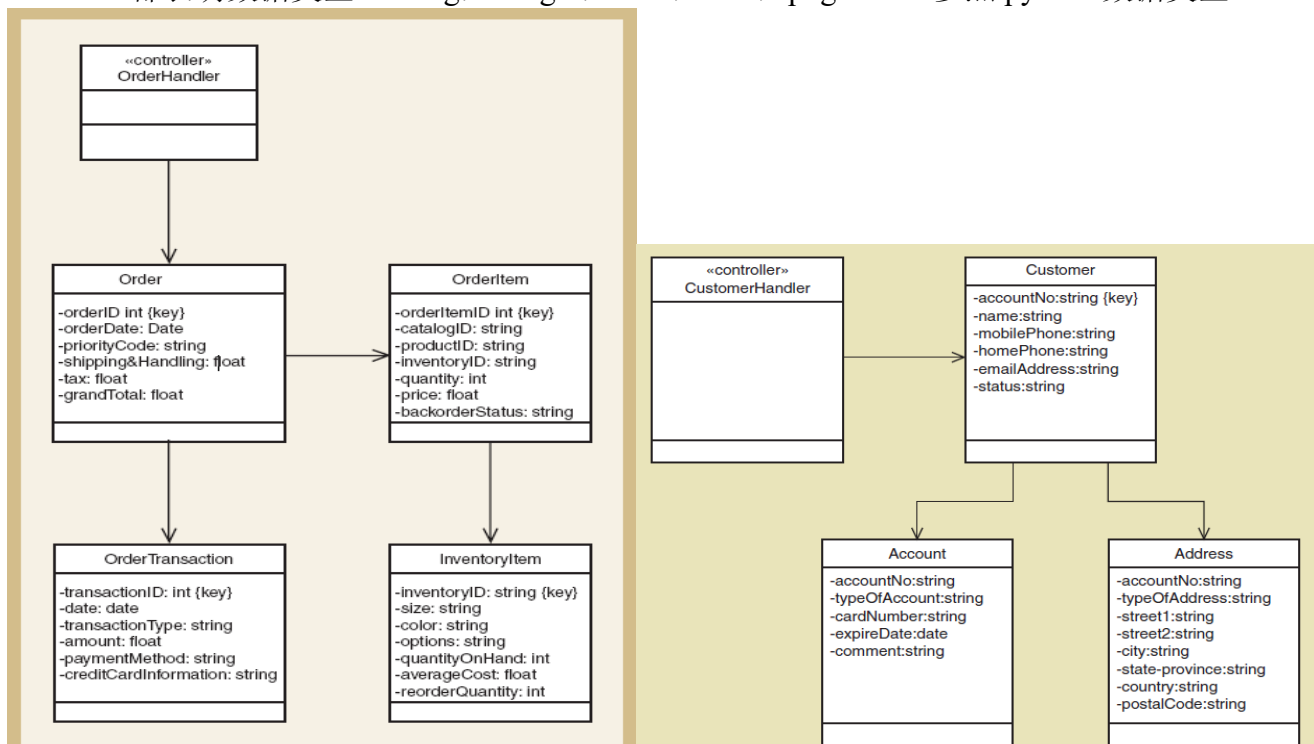
所有 attribute 按照 camelback 格式 (从第二个单词及后面的首字母大写, 不空格)

Attribute 前面都是“-”减号, 表示 visibility 的类型是 private, 写第二行, first-cut DCD 第三行空着

(在 complete DCD 中, 所有 **method 命令的前面都是“+”加号**, visibility 类型是 public, 写第三行)

不需要体现子类, 都当成普通 class, 不需要标出关系, 只要箭头 (navigation)

Attribute 都表明数据类型: string, integer, float, date, png…… (参照 python 数据类型)



Design System Sequence Diagram (DSSD)

根据 SSD 来画 DSSD

需要体现 3 个 layer——View layer, domain layer=business logic layer, and data layer

User 和 XXXhandler 之间的输入输出和 SSD 一样，并且中间加个 View layer，名字自取
所涉及到的 object 都是 system，都需要冒号下划线

Object 和谁相关，就从谁处 create

查找/添加/更改/删除 attribute，需要 object 与相应 objectDA 互动（DA 是 data access，不是 data）

所有需要返回给 user 的输出都要抵达 handler

如果不返回给 user，只保留 object 和相应 DA 的互动即可，DA 给 object 输出为 confirmation

检查 loop、opt、alt 结构，用实线分隔，子情况中的输入输出最好成对出现

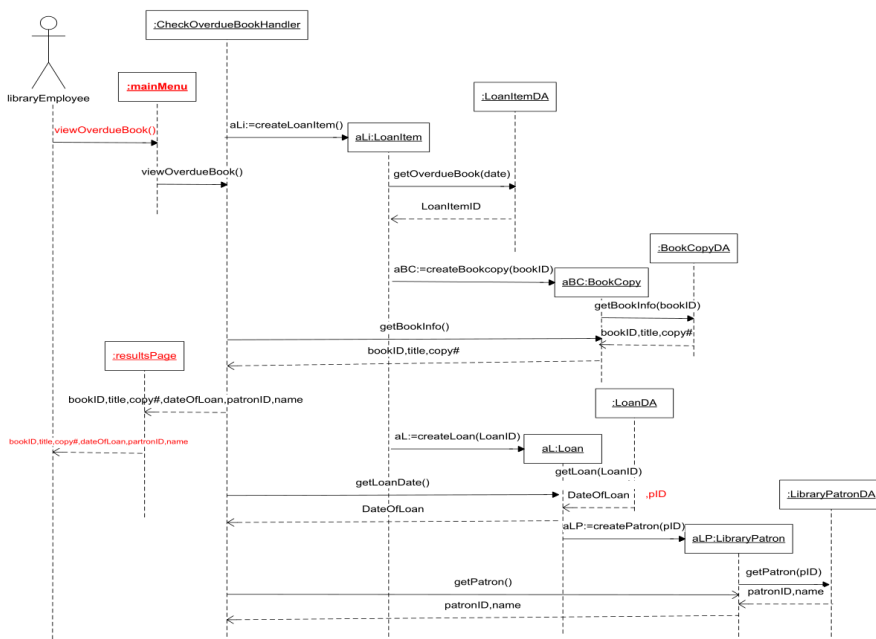
命令的动词随使用，不固定

Object 下面的竖线都用虚线也行，加个长方形条也行

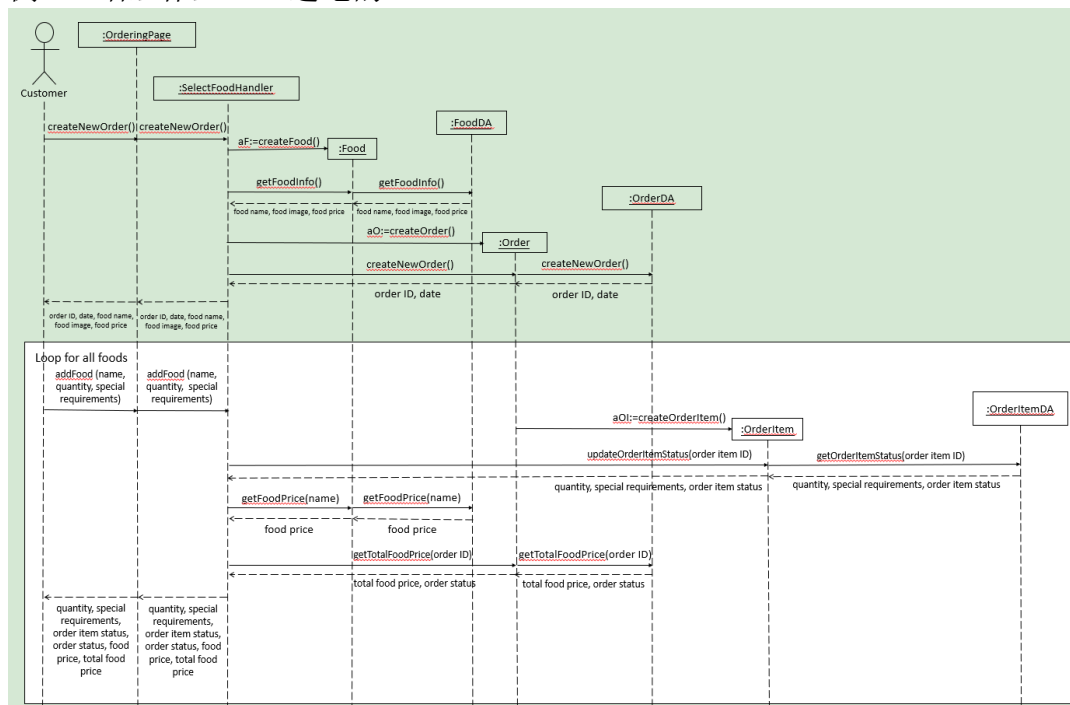
输入输出的完整画法、简便画法都可以，随便

输入输出线的上下关系体现发生的先后顺序

例 1：查过期书



例 2：咱组作业——选吃的



Complete Design Class Diagram (Complete DCD)

补全 first-cut DCD

补什么？第三行

根据 DSSD 中与 object 有关的所有 method 指令，与 DA 有关的不要

比如：

createLoan(loan ID)、getLoanDate()

createFood()、getFoodInfo()、getFoodPrice()

指令中括号里的信息也要

-写在 class 的 attribute 前，放 class 第二行

+写在 complete DCD 的 method 指令前，放 class 第三行

